

CSE 2 • S5 MINOR PROJECT • 0TH REVIEW

AI-Powered Smart Expense Tracker

Automated parsing, AI categorization, and intelligent budgeting for personal finance.

TEAM

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GUIDE

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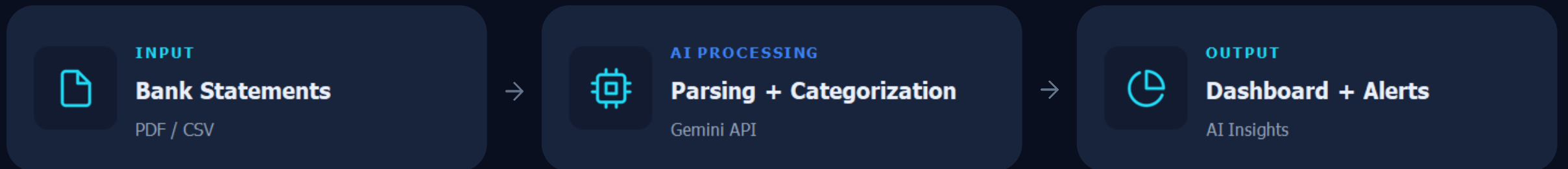
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Abstract

A web app where a user uploads their bank statement, and the app automatically reads every transaction, uses AI to figure out what category each one belongs to (Food, Bills, Shopping, etc.), and shows it all on a dashboard with charts and insights. Unlike existing trackers that require manual entry or only report raw data, our system automates the entire process and adds proactive guidance, such as, budget alerts, AI savings suggestions and user-correctable categorization, while remaining completely free and open.

In Short:



Objectives

01

Automate

Automate extraction and categorization of transactions from bank statements (PDF/CSV) using AI



02

Secure

Provide a secure, user-specific dashboard for tracking income, expenses and spending trends



03

Monitor

Enable budget limit setting with real-time alerts across overall and per-category spending



04

Optimize

Generate AI-driven spending insights and personalized savings suggestions



Existing System



Manual Tracking Apps

Manual expense tracking apps require users to enter every transaction by hand



Bank Apps

Bank apps show raw transaction lists without meaningful categorization or insights



Spreadsheet Tracking

Spreadsheet-based tracking is time-consuming and prone to errors



Existing Tools

Most existing tools lack automated statement parsing or AI-based categorization

Disadvantages

01 High Manual Effort



High manual effort discourages consistent tracking, leading to inaccurate records

02 No Automatic Categorization



No automatic categorization means users must label every transaction themselves

03 Limited Personalization



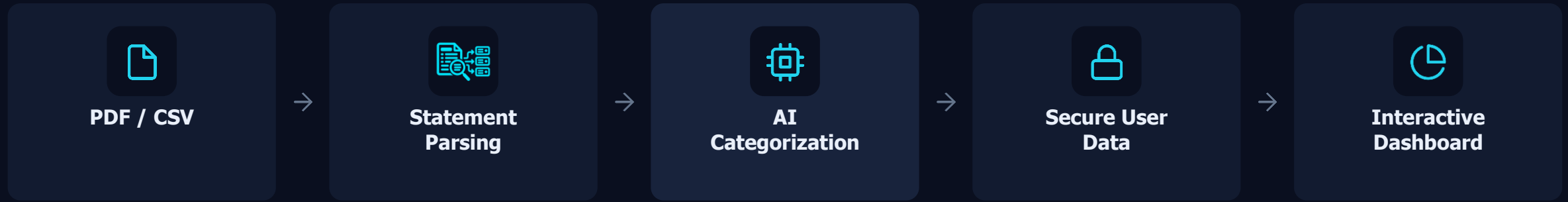
Little to no personalized insight into spending behavior or savings opportunities

04 No Proactive Alerts



Absence of proactive budget alerts allows overspending to go unnoticed

Proposed System



PDFBox · Apache Commons CSV · Google Gemini API · JWT · Dashboard

01

Upload bank statements (PDF/CSV) and auto-parse transactions using PDFBox and Apache Commons CSV

02

Use the Google Gemini API to automatically categorize transactions and identify merchants

03

Secure JWT-based authentication with per-user data isolation

04

Interactive dashboard with charts, budget alerts, and AI-generated savings tips

Advantages



Automation

Eliminates manual data entry through automated statement parsing



Accuracy

AI categorization saves time and improves accuracy over manual tagging



Proactive Budgeting

Real-time budget alerts help users control overspending proactively



Visual Analytics

Visual analytics (pie charts, trend charts) make spending patterns easy to understand

Hardware and Software Specifications



Hardware

Standard PC / laptop
Minimum 4GB RAM
Stable internet connection



Backend

Java 21
Spring Boot 3.5
MySQL



Frontend

React
Tailwind CSS
Recharts



APIs and Libraries

Google Gemini API
Apache PDFBox
Apache Commons CSV



Development Tools

IntelliJ IDEA
VS Code
Postman, Git

Methodology

01

Requirement Analysis

Requirement analysis and technology stack finalization

02

Backend Development

Backend development: entity design, JWT authentication, transaction CRUD APIs

03

Statement Parsing

Statement parsing module: PDF and CSV parsers converging into a unified transaction format

04

AI Integration

AI integration: Gemini API for categorization, merchant extraction, and savings suggestions

05

Frontend Development

Frontend development: dashboard, analytics, budget alerts and insights UI

06

Testing & Refinement

End-to-end testing and iterative refinement of each module

Tools Used



Backend



Java



Spring Boot



MySQL

 Maven

Frontend



React



Tailwind CSS



Recharts



Axios



AI & Parsing



Google Gemini API



Apache PDFBox



Apache Commons CSV



Development



IntelliJ IDEA



VS Code



Postman



Git / GitHub

Expected Outcomes



Functional Application

A fully functional web application that parses bank statements and auto-categorizes transactions



Real-Time Dashboard

A real-time dashboard showing income, expenses, savings and category-wise spending



Alerts & AI Suggestions

Automated budget alerts and AI-generated personalized savings suggestions



Portfolio-Ready System

A deployable, portfolio-ready full-stack application demonstrating practical AI integration

Future Enhancements



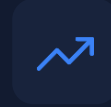
More Formats & Currencies

Support for more bank statement formats and multiple currencies



Mobile Application

A mobile application version for on-the-go expense tracking



Predictive Analytics

Predictive analytics to forecast future spending and savings goals



Bank API Integration

Direct integration with bank APIs for real-time transaction syncing

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Thank You.

Questions and feedback are welcome.

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Difference



	Expense AI	FinArt	Other AI Trackers	Ours
Cost	Paid (₹260 / week)	Free	Free (mostly)	Free
Input Method	Manual entry, receipts, chat	SMS-based (automatic)	Manual entry, voice, or chat	Upload full bank statement (CSV/PDF)
Historical Data	Limited to logged entries	Only from install date onward	None — starts from first log	Entire past history in one upload
Categorization	AI-based	Rule-based (SMS parsing)	AI, on user-typed text	AI (Gemini), on real transaction data